"The Statistics of Sharpe Ratios": A Comment

In the article "The Statistics of Sharpe Ratios" by Andrew Lo (July/August 2002), the author claims that under the assumptions that returns are identically and independently distributed and have finite mean μ and variance σ^2 and based on the Central Limit Theorem, the asymptotic distribution of the (centered and scaled) sample variance, $\sqrt{T}\left(\hat{\sigma}^2 - \sigma^2\right)$, is given by $N(0, 2\sigma^4)$; see Equation 5.

This claim, however, is true only under the *additional* assumption that the returns have a normal distribution. In general, the asymptotic variance depends also on the kurtosis (the fatness of the tails) of the underlying distribution.

This point is more than an academic detail; it has practical importance. As is well known, the returns of many assets have a distribution with fatter tails than the normal distribution has. In such cases, the number $2\sigma^4$ understates the true asymptotic variance—possibly by a lot. As a result, the confidence interval for the Sharpe ratio presented in Equation 10 will typically be too narrow, which will give the user of the interval a false sense of security.

A more general formula that would give trustworthy inference even if the return distribution is not normal would be quite straightforward to derive. Perhaps the record can be set straight?

Michael Wolf
Universitat Pompeu Fabra
Barcelona, Spain

"The Statistics of Sharpe Ratios": Author's Response

I thank Professor Wolf for pointing out a potentially important qualification in my derivation of the asymptotic distribution for the Sharpe ratio under the assumption of independently and identically distributed (IID) returns. As it is written, the asymptotic distribution for the variance estimator $\hat{\sigma}^2$ does indeed require the assumption of normality. Without normality, the asymptotic distribution of $\hat{\sigma}^2$ is given by

$$\left(\sqrt{T}\hat{\sigma}^2 - \sigma^2\right) \stackrel{a}{\sim} N(0, \theta), \tag{R1}$$

$$\theta = E\left[(R_t - \mu)^4\right] - \sigma^4, \tag{R2}$$

and theta can be estimated in the natural way:

$$\hat{\theta} = \frac{1}{T}\sum_{t=1}^{T}(R_t - \hat{\mu})^4 - \hat{\sigma}^4, \tag{R3}$$

where $\hat{\mu}$ and $\hat{\sigma}$ are the usual sample mean and variance estimators. Using Equation R1 in place of $2\sigma^4$ in the derivation of the Sharpe ratio's asymptotic distribution (see Appendix A of "The Statistics of Sharpe Ratios") will then yield the correct result for general IID symmetrically distributed returns with finite fourth moments. When R_t is normally distributed, it is well known that

$$E[(R_t - \mu)^4] = 3\sigma^4; \tag{R4}$$

hence, θ reduces to $2\sigma^4$ in this case, as Professor Wolf observes.

It should be emphasized, however, that the IID case was meant primarily to be illustrative and is only of limited practical value because the IID assumption is often violated for financial data. In particular, asset returns are rarely identically distributed, and statistically significant forms of dependence have been documented in many contexts (see, for example, Lo and MacKinlay 1999 and Getmansky, Lo, and Makarov, forthcoming 2003). Therefore, the generalized method of moments estimator proposed in the "Non-IID" section of "The Statistics of Sharpe Ratios" should be used instead, and those results do apply to nonnormal asset returns with quite general forms of dependence.

Andrew W. Lo
Sloan School of Management
Massachusetts Institute of Technology

References

Getmansky, M., A. Lo, and I. Makarov. Forthcoming 2003. "An Econometric Analysis of Serial Correlation and Illiquidity in Hedge-Fund Returns." *Journal of Financial Economics*.

Lo, A., and C. MacKinlay. 1999. *A Non-Random Walk Down Wall Street*. Princeton, NJ: Princeton University Press.

The views expressed in Letters to the Editor are those of the authors, not of AIMR or the Financial Analysts Journal.